

# Towards Adding Pseudo-Haptic Effects to Acoustic Levitation Displays

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## ABSTRACT

Acoustic levitation displays are a novel technology that use ultrasound to levitate small ‘particles’ in mid-air, to create physical 3D content. Interaction with levitated content is currently limited to simple operations, e.g., moving a particle. In this paper, we investigate the addition of pseudo-haptic effects to levitation displays, to enrich the interaction with new sensory experiences and provide simulated touch feedback. We describe a range of pseudo-haptic effects and possible implementations, based on the relationship between user control and particle response. A user-study is proposed to evaluate the potential contribution of pseudo-haptic effects when used with acoustic levitation displays.

## CCS CONCEPTS

• Human-centered computing Haptic devices; • Human-centered computing Gestural input;

## KEYWORDS

Acoustic levitation displays, pseudo-haptic effects, interaction techniques, particle displays, physical pixels.

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## 1 INTRODUCTION

Acoustic levitation displays levitate small physical objects in air using an array of ultrasound emitters, allowing content composed from physical ‘particles’ [8]. Acoustic pressure patterns “trap” particles (typically small polystyrene beads) in precisely defined locations. Multiple traps can be created and moved precisely and independently, by manipulating the phase signature of the ultrasound array [23, 26]. As in Ishii’s *radical atoms* concept [12], these particles can be seen as atomic elements of content to build more complex objects users can interact with (Figure 1). Alongside developments to improve the levitation technology itself, there are many questions

about how to interact with the levitated content. Several studies have investigated interaction techniques, e.g., for selecting [7] and moving particles [2] and avoiding collisions during movement [28]. Others have explored novel presentation techniques, e.g., combining static and dynamic physical materials [6, 24, 25], or using high speed motion for persistence of vision graphics [11].

Our focus is on improving feedback during contactless gesture interaction, using pseudo-haptic effects. Many studies have shown that haptic feedback can improve interaction, from desktop GUIs [30], to phones [4], in-car interfaces [20] and even Air Traffic Control [27, 29]. Ultrasound arrays themselves can be used to provide haptic feedback [3]. However, this is not straightforward because acoustic reflections from a hand interacting near the display can cause instability in the levitated particles. Therefore, keeping the hands further away from the levitation array is important. Our novel approach is to use pseudo-haptic effects instead, “*a technique for simulating haptic sensations [...] capitalizing on the properties of human perception, [which] uses visual feedback and verges on sensory illusion*” [21], to enrich interaction with haptic cues.

Pseudo-haptic effects rely on visuo-haptic integration [9]: by changing the Control/Display (C/D) gain dynamically, different haptic effects can be simulated [21]. Designers can use this type of illusion to produce sensations that can be perceived without producing any mechanical forces on the body. This principle has been described for texture and friction [1, 15, 18], stickiness [22], mass/inertia [5, 13], and stiffness [14, 16, 17]. Pseudo-haptic effects are commonly applied to dragging operations with mouse pointers [5, 18, 21]. There has been no research so far looking at if, and how, such effects can be used with levitating objects, and how they should be designed to work with levitated particles.

Acoustic levitation displays are becoming more and more usable for concrete applications, close to what one can imagine doing with holograms [11]. Adding haptics capabilities to this technology without making it more complex can enrich it significantly. Realistic haptic feedback and effects could thus be envisaged and could significantly improve the interaction with the levitating content.

In this paper, we present ongoing research into a variety of pseudo-haptic effects for particle movement tasks in acoustic levitation displays. Initial investigations led us to create preliminary designs for four types of effects: *Friction*, *Sliding*, *Texture*, and *Stickiness*, with variations of each type. We also introduce additional effects: *Inertia* to render the mass of an object and *Stiffness* to show compliance or deformation. We have implemented the first four effects in our levitation device; the next stage will be to test their effectiveness (unfortunately our lab has been closed for a significant time due to COVID restrictions).

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Figure 1: An example of eight polystyrene beads positioned to form a cube in mid-air, a *compound object* with many particles. Ultrasound standing waves are produced by emitters located above and below the particles.

## 2 PSEUDO-HAPTIC EFFECTS FOR ACOUSTIC LEVITATION DISPLAYS

In this section, we describe our design contributions for the pseudo-haptic effects of *Friction*, *Sliding*, *Textures* and *Stickiness*. Implementations are given, focusing on the manipulation of C/D gain. *Inertia* and *Stiffness* are discussed as well.

### 2.1 Friction

The sensation of friction is an important component while exploring our surrounding environment and contributes to the perception of texture. Friction is “*the production of force as a result of relative movement between two objects in contact*” [10], which occurs every time our skin is in contact with an object. Pseudo-haptic friction illusions can improve mouse pointer interaction with screens [19] and could potentially improve interaction with acoustic levitation displays using similar techniques, in addition to supporting richer interaction through perceptual illusions of texture.

When dragging an object (in our case a levitating particle) a pseudo-haptic *Friction* effect would produce movement resistance, dynamically reducing velocity and creating a discrepancy between the amplitude of the drag gesture (longer) and the distance travelled by the particle (smaller). The larger the discrepancy, the greater the friction effect. It could be perceived as the particle’s velocity becomes lower than the control gesture. This effect is produced by reducing the C/D gain, the ratio of control to display movement.

Our first implementation (Type-1) was inspired by [19], where users had to slide a virtual cube over a textured surface. The cube decelerated when passing over this surface, by reducing the C/D gain. We implemented this effect for an acoustically levitated particle in the same way, by decreasing the C/D gain when the particle is dragged across a predefined area. Acoustic levitation displays place constraints on particle movement speed: if movements are too fast then particles may fall out of the display. We normally use a C/D gain of 0.35 with our levitation display for robust rapid user movements (although this is specific to our hardware).

Our second implementation (Type-2) is built on the same principle, however the C/D gain value for slowing the movement is not constant but randomised. In everyday life, when we touch a



Figure 2: A use case of a pseudo-haptic effect of Friction: the parts in red delimit the areas in which the effect (Type-1 or 2) is enabled. The user has to guide the levitating particle (white dot) through an invisible maze. The path is not known to the user; no visual cues are provided.

surface, the opposing force is not always constant and the finger moves in bursts. We informally tested it during the development process with a value for the C/D gain randomly generated between 0.1 and 0.35 (our default value) for providing the users with a more textured friction effect, less linear than the first implementation.

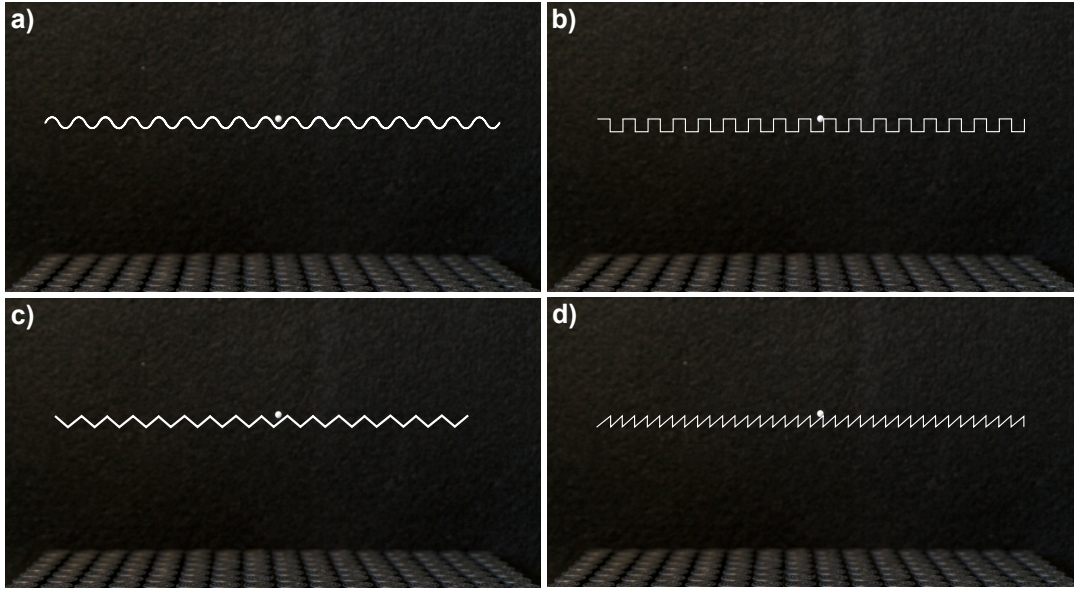
To exemplify the potential added-value of a *Friction* effect, imagine a game in which users would have to find a path within an invisible maze by dragging a levitated particle. There is only one path to reach the exit and no visual cue are provided to find this path. A *Friction* effect could be useful to indicate the users the border of the path (Figure 2). Similar effects could provide guidance during input tasks. This game will be considered for a user study with the objective of finding (among Types 1 and 2) the best way to guide the user from a point A to a point B during a dragging operation over a levitating particle.

### 2.2 Sliding

*Sliding* effects are achieved in a similar way to *Friction*. However, the C/D gain is increased instead of decreased within a predefined area, as in [19]. This principle generates a positive shift in user’s gestures speed relative to that of the levitating particle. A sliding sensation (the opposite of friction) is then experienced. During the development phase, we evaluated such Sliding effects and using a C/D gain up to 0.6 over a very short distance (i.e., 1–2 cm) while avoiding levitating particle instability. It would also appear that beyond this value, and over a distance of approximately 2 cm at most, a greater C/D gain is not possible with our hardware because of a high risk levitating particles being ejected. However, these values still need to be refined experimentally to offer users the most convincing and stable sliding effect possible.

### 2.3 Texture

*Friction* and *Sliding* are first steps towards enriching levitating particles with more complex pseudo-haptic sensations of *Texture*. However, simply offsetting the C/D gain may not be sufficient for creating a convincing experience of moving across a textured surface. We believe that adding a spatial dimension to particle



**Figure 3: Different transfer functions for oscillating a moving particle could result in different pseudo-haptic texture sensations. Sine (a), square (b), triangle (c) and sawtooth (d) wave functions are four examples, which could be combined with corresponding change in C/D gain as the particle (white circle) moves across the “bumps” and “holes”.**

movement may create a more convincing illusion of texture. This can be achieved by manipulating the position of the levitating particle in addition to manipulating C/D gain .

Texture has been widely studied in haptics research, since successfully producing textures would allow haptic designers to create virtual objects with convincing material properties. Lécuyer, Burkhardt & Étienne [18] and Argelaguet *et al.* [1] have described techniques for creating pseudo-haptic experiences of texture when using a mouse pointer on a flat screen. Their technique manipulates the movement of the mouse pointer, so that it looks like the pointer is moving over a textured surface. Lécuyer, Burkhardt & Étienne give the example of moving the pointer over “bumps and holes”: the pointer speeds up as it falls into a hole, then slows down as it rises out of the other side; likewise, the pointer slows down as it moves over a bump, then speeds up as it falls off the other side. Their results show that this technique was successful at creating the haptic experience of moving the mouse over a textured surface. Moreover, they obtained the same results without using a background image that would give the user visual insights on pointer trajectory, and showed that only pointer movement and velocity were sufficient to trigger the haptic texture illusion.

When interaction involves visual content, in our case levitated particles, pseudo-haptic effects could be created by manipulating the position of the levitating particle as it moves, introducing a discrepancy with the control gesture trajectory. The movement of a particle along a path composed of oscillating “bumps and holes”, with corresponding variance in C/D gain , may replicate this pseudo-haptic sensation. Different textures could be simulated by manipulating the period and amplitude of the oscillation, and the transfer function used to vary the particle position relative to its trajectory. For example, Figure 3 shows four examples of transfer

functions that could potentially be used to create different textured sensations, although these require investigation. Similarly, a noise function will also be evaluated to see if that affects a perceived sensation of *Texture*.

## 2.4 Stickiness

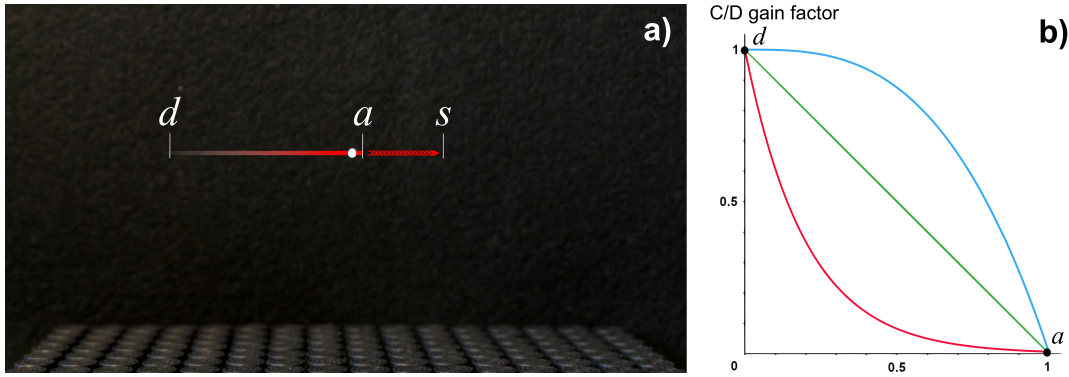
*Stickiness* [21] is another interesting pseudo-haptic effect. It has previously been found useful when manipulating on-screen content with pointing devices, e.g., organising windows in a graphical environment [22]. Our implementation of *Stickiness* decelerates the user-controlled levitating particle starting from a predefined point in the levitation space (see point *d* on Figure 4–a) by manipulating the C/D gain . This deceleration is controlled until a second point *a*, where the particle will start to be attracted to a third point *s* (‘sticking point’). Users have no control over the particle by the time it is attracted to this sticking point (i.e., between *a* and *s*).

We implemented three different types of deceleration to explore a convincing way to slow the particle and, consequently, to provide the users with an understandable effect. Three deceleration functions (linear, cubic, exponential, as shown in Figure 4–b) have been implemented, to map particle’s *x*-position to C/D gain :

$$\begin{aligned} \text{lin\_dec}(x) &= 1 - x, \\ \text{cub\_dec}(x) &= 1 - x^3, \\ \text{exp\_dec}(x) &= \frac{1}{e^{5x}}. \end{aligned} \tag{1}$$

## 2.5 Mass, Inertia

Haptic sensations like *Friction*, *Sliding*, *Texture* and *Stickiness* are surface properties. More complex haptic sensations, representing the mass and physical characteristics of a virtual object, may also be



**Figure 4: (a) Stickiness effect principle.** The particle (white point) starts to decelerate at point  $d$  according to a deceleration function (see (b)).  $a$  is the point where the particle velocity is at its lowest and where the particle starts to be attracted towards  $s$ , which is where the particle is eventually stuck. Distances shown ( $d \rightarrow a$  and  $a \rightarrow s$ ) are only visual examples and will need to be further specified; (b) Graphs for the deceleration functions  $\text{lin\_dec}$ ,  $\text{cub\_dec}$  and  $\text{exp\_dec}$  representing linear, cubic and exponential deceleration (respectively in green, blue and red; see Equation 1).  $x$ -axis represents the normalised value of a particle's  $x$  position;  $y$ -axis represents a multiplying factor applied to the C/D gain .

possible. The *Inertia* pseudo-haptic effect creates the illusion of mass for a levitated particle when it is moved, which has already been studied for objects in VR [31]. This could potentially be achieved for acoustic levitation displays by considering a moment of inertia for a particle, which then takes the role of a rigid body. By manipulating the velocity of the particle, successive positions can be computed for a levitating particle to create a lag between a user's gesture and its movement path. This lag should be proportional to user's fingertip velocity: the 'heavier' the particle, the longer the lag. That way, the particle will start moving after user's gesture has started and will stop after user's gesture has ended. Our hypothesis is that it will visually create a feeling of both slowing down and heaviness to create the illusion of inertia.

Such a pseudo-haptic effect is interesting for acoustic levitation displays for two main reasons. First, it is a means to enrich the levitating content with physical characteristics, by adding a sense of weight to these untouchable objects. Incorporating aspects like inertia may be a good way to make interaction with levitation technology more convincing because it will provide users with physical metaphors to help understand and predict particle behaviours during control gestures. Secondly, *mass* can be seen as an additional dimension to provide the users with information of weight. Assigning mass to objects can be a way to assign importance or rank to them, for example.

## 2.6 Stiffness

*Stiffness* can provide a sense of *compliance* and *deformation* to an object. Compliance and deformation can be seen as the two extreme values of a gradual 'stiffness scale' for rigidity. These effects are different from the others described above in that must be applied to an object that can be visually deformed. Since a single levitating particle cannot be deformed, this effect requires compound objects consisting of multiple levitating particles. Otherwise, no visual reference would be provided and a sense of deformation (consequently, a sense of stiffness) could not be perceived.

Implementation is more complex than the other effects because of novel parameters. The first parameter is the *type* of deformation, i.e., how interaction would impact the shape of the levitating content. Depending on their nature, everyday objects react differently to a stress that deforms them. Initially, this parameter could be simplified by considering only one type of deformation (for example, a contraction of the shape formed by several levitating particles, see Figure 1). The second parameter is *deformation factor*, i.e., considering the speed of user's gestures, the rate at which the levitating object is impacted. The third parameter is *diffusion factor*, i.e., how far does the deformation propagate in the impacted object: it is only deformed around the point of impact, or over its entirety?

## 2.7 Application to Compound Objects

With the exception of *Stiffness*, the envisioned pseudo-haptic effects described previously can be applied to a single levitating particle. The outlined implementations extend previous works on 2D pointer effects, to be applied to physical objects that levitate and move in a 3D volume. One goal of this research is to eventually apply these effects to compound objects formed from multiple particles, such as the cube in Figure 1. The single particle instances of *Friction*, *Sliding*, *Texture*, *Stickiness* and *Inertia* could be extended to manipulate each particle in a compound object.

Considering, as an example, the cube in Figure 1, all the pseudo-haptic effects described in this paper would simply be applied to each of the eight particles composing the object. For *Inertia* effect, a mass can even be associated to the manipulated compound object by simply considering a value that is proportional to the number of particles composing it; i.e., the more particles a compound object contains, the heavier it may be perceived. However, generalising to compound objects should be properly investigated, because the inherent instability of levitation is proportional to the number of levitating particles (typically more stable when there are fewer particles). We plan to study this in future work, after first investigating pseudo-haptic perception with a single particle.

### 3 PROGRESS & TARGETED EXPERIMENTS

We have informally evaluated these effects during their development, e.g., by exploring appropriate parameters and observing their effects while controlling a levitating particle. This has given us initial working prototypes in our levitation device (two 16x16 arrays of 40 kHz emitters, 23 cm apart). Our next step is to formally evaluate our implementations for further refinement. A series of short user-studies is planned to evaluate perception and convincingness of each of the effects. Participants will interact with our prototype by directly controlling a levitating particle using mid-air gestures in controlled movement tasks. We will collect mixed-methods data to interrogate the extent to which each pseudo-haptic effect is perceived.

Key parameters will be controlled to allow us to find the values that generate the best perception of haptic feedback: *C/D gain values* for Friction and Sliding, the *type of signal*, the *amplitude* and *frequency* for Texture,  $d \rightarrow a$  and  $a \rightarrow s$  *distances* and the *decreasing function* for Stickiness. Experimental variables for Inertia and Stiffness effects will be defined in later work.

The outlined user study will address important aspects of pseudo-haptic perception using these techniques, such as user's perception of reproduced physical behaviours. It is not yet straightforward to distinguish between effects like friction and inertia: the beginning of a dragging operation on a heavy object could make the user perceive friction; however, mass should not be perceived as such. Implementing inertia to closely replicate physical laws could be a solution, and testing it with users could answer this design issue. Our implementations still require refinement and our user study can inform this.

### 4 CONCLUSION

This paper describes our initial design and implementation of a range of pseudo-haptic effects for acoustic levitation displays based on the manipulation of C/D gain and/or particle position. These effects have been studied with other types of display, such as VR and mouse pointer GUIs, but have not yet been tested for levitated content in a see-through physical display medium. For each effect, we outlined our basic design approach and our initial implementations based on informal formative evaluation. The next step in this work is to conduct a user evaluation to investigate perception and establish reliable parameters for evoking these perceived haptic sensations. Pseudo-haptic effects could be beneficial for levitation displays as they would allow user interface designers to design richer interactions without adding more hardware or incorporating more visible content into the display volume. These effects could pave the way to more usable interactions with levitating content.

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### REFERENCES

- [1] Ferran Argelaguet, David Antonio Gómez Jáuregui, Maud Marchal, and Anatole Lécuyer. 2012. A Novel Approach for Pseudo-haptic Textures Based on Curvature Information. In *Haptics: Perception, Devices, Mobility, and Communication*, Poika Isokoski and Jukka Springare (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 1–12.
- [2] Myroslav Bachynskyi, Viktorija Paneva, and Jörg Müller. 2018. LeviCursor: Dexterous Interaction with a Levitating Object. In *Proceedings of the 2018 ACM International Conference on Interactive Surfaces and Spaces* (Tokyo, Japan) (ISS '18). Association for Computing Machinery, New York, NY, USA, 253–262. <https://doi.org/10.1145/3279778.3279802>
- [3] Tom Carter, Sue Ann Seah, Benjamin Long, Bruce Drinkwater, and Sriram Subramanian. 2013. UltraHaptics: Multi-Point Mid-Air Haptic Feedback for Touch Surfaces. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) (UIST '13). Association for Computing Machinery, New York, NY, USA, 505–514. <https://doi.org/10.1145/2501988.2502018>
- [4] Angela Chang and Conor O'Sullivan. 2005. Audio-Haptic Feedback in Mobile Phones. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems* (Portland, OR, USA) (CHI EA '05). Association for Computing Machinery, New York, NY, USA, 1264–1267. <https://doi.org/10.1145/1056808.1056892>
- [5] L. Dominjon, A. Lecuyer, J. Burkhardt, P. Richard, and S. Richir. 2005. Influence of control/display ratio on the perception of mass of manipulated objects in virtual environments. In *IEEE Proceedings. VR 2005. Virtual Reality*, 2005. 19–25. <https://doi.org/10.1109/VR.2005.1492749>
- [6] Euan Freeman, Asier Marzo, Praxitelis B. Kourtelos, Julie R. Williamson, and Stephen Brewster. 2019. Enhancing Physical Objects with Actuated Levitating Particles. In *Proceedings of the 8th ACM International Symposium on Pervasive Displays* (Palermo, Italy) (PerDis '19). Association for Computing Machinery, New York, NY, USA, Article 2, 7 pages. <https://doi.org/10.1145/3321335.3324939>
- [7] Euan Freeman, Julie Williamson, Sriram Subramanian, and Stephen Brewster. 2018. Point-and-Shake: Selecting from Levitating Object Displays. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3173574.3173592>
- [8] Euan Freeman, Julie R. Williamson, Praxitelis Kourtelos, and Stephen Brewster. 2018. Levitating Particle Displays with Interactive Voxels. In *Proceedings of the 7th ACM International Symposium on Pervasive Displays - PerDis '18*. ACM Press, Article 15. <https://doi.org/10.1145/3205873.3205878>
- [9] Steve Guest and Charles Spence. 2003. Tactile dominance in speeded discrimination of textures. *Experimental brain research* 150, 2 (2003), 201–207.
- [10] Vincent Hayward and Brian Armstrong. 2000. A new computational model of friction applied to haptic rendering. In *Experimental Robotics VI*. Springer London, London, 403–412.
- [11] Ryuji Hirayama, Diego Martinez Plasencia, Nobuyuki Masuda, and Sriram Subramanian. 2019. A volumetric display for visual, tactile and audio presentation using acoustic trapping. *Nature* 575, 7782 (2019), 320–323.
- [12] Hiroshi Ishii, Dávid Lakatos, Leonardo Bonanni, and Jean-Baptiste Labrune. 2012. Radical Atoms: Beyond Tangible Bits, toward Transformable Materials. *Interactions* 19, 1 (Jan. 2012), 38–51. <https://doi.org/10.1145/2065327.2065337>
- [13] Topi Kaaresoja, Eve Hoggan, and Emilia Anttila. 2011. Playing with Tactile Feedback Latency in Touchscreen Interaction: Two Approaches. In *Human-Computer Interaction – INTERACT 2011*, Pedro Campos, Nicholas Graham, Joaquim Jorge, Nuno Nunes, Philippe Palanque, and Marco Winckler (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 554–571.
- [14] Johan Kildal. 2010. 3D-Press: Haptic Illusion of Compliance When Pressing on a Rigid Surface. In *International Conference on Multimodal Interfaces and the Workshop on Machine Learning for Multimodal Interaction* (Beijing, China) (ICMI-MLMI '10). Association for Computing Machinery, New York, NY, USA, Article 21, 8 pages. <https://doi.org/10.1145/1891903.1891931>
- [15] Johan Kildal. 2011. Evoked Friction on a Smooth Touch Device. In *CHI '11 Extended Abstracts on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI EA '11). Association for Computing Machinery, New York, NY, USA, 2233–2238. <https://doi.org/10.1145/1979742.1979939>
- [16] Johan Kildal. 2012. Kooboh: Variable Tangible Properties in a Handheld Haptic-Illusion Box. In *Haptics: Perception, Devices, Mobility, and Communication*, Poika Isokoski and Jukka Springare (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 191–194.
- [17] Chi-Hsia Lai, Matti Niinimäki, Koray Tahiroglu, Johan Kildal, and Teemu Ahmaniemi. 2011. Perceived Physicality in Audio-Enhanced Force Input. In *Proceedings of the 13th International Conference on Multimodal Interfaces* (Alicante, Spain) (ICMI '11). Association for Computing Machinery, New York, NY, USA, 287–294. <https://doi.org/10.1145/2070481.2070533>
- [18] Anatole Lécuyer, Jean-Marie Burkhardt, and Laurent Etienne. 2004. Feeling Bumps and Holes without a Haptic Interface: The Perception of Pseudo-Haptic Textures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) (CHI '04). Association for Computing Machinery, New York, NY, USA, 239–246. <https://doi.org/10.1145/985692.985723>
- [19] A. Lecuyer, S. Coquillart, A. Kheddar, P. Richard, and P. Coiffet. 2000. Pseudo-haptic feedback: can isometric input devices simulate force feedback?. In *Proceedings IEEE Virtual Reality 2000* (Cat. No.00CB37048). 83–90. <https://doi.org/10.1109/VR.2000.840369>
- [20] John D. Lee, Joshua D. Hoffman, and Elizabeth Hayes. 2004. Collision Warning Design to Mitigate Driver Distraction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) (CHI '04). Association

- for Computing Machinery, New York, NY, USA, 65–72. <https://doi.org/10.1145/985692.985701>
- [21] Anatole Lécuyer. 2009. Simulating Haptic Feedback Using Vision: A Survey of Research and Applications of Pseudo-Haptic Feedback. *Presence: Teleoperators and Virtual Environments* 18, 1 (2009), 39–53. <https://doi.org/10.1162/pres.18.1.39>
- [22] Regan L. Mandryk, Malcolm E. Rodgers, and Kori M. Inkpen. 2005. Sticky Widgets: Pseudo-Haptic Widget Enhancements for Multi-Monitor Displays. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems* (Portland, OR, USA) (CHI EA '05). Association for Computing Machinery, New York, NY, USA, 1621–1624. <https://doi.org/10.1145/1056808.1056981>
- [23] Asier Marzo, Sue Ann Seah, Bruce W. Drinkwater, Deepak Ranjan Sahoo, Benjamin Long, and Sriram Subramanian. 2015. Holographic acoustic elements for manipulation of levitated objects. *Nature Communications* 6 (2015), Article 8661. <https://doi.org/10.1038/ncomms9661>
- [24] Rafael Morales, Asier Marzo, Sriram Subramanian, and Diego Martínez. 2019. LeviProps: Animating Levitated Optimized Fabric Structures Using Holographic Acoustic Tweezers. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 651–661. <https://doi.org/10.1145/3332165.3347882>
- [25] Rafael Morales González, Euan Freeman, and Orestis Georgiou. 2020. Levi-Loop: A Mid-Air Gesture Controlled Levitating Particle Game. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3334480.3383152>
- [26] Themis Omirou, Asier Marzo, Sue Ann Seah, and Sriram Subramanian. 2015. LeviPath: Modular Acoustic Levitation for 3D Path Visualisations. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems - CHI '15*. ACM Press, 309–312. <https://doi.org/10.1145/2702123.2702333>
- [27] Maxime Reynal, Pietro Aricò, Jean-Paul Imbert, Christophe Hurter, Gianluca Borghini, Gianluca Di Flumeri, Nicolina Sciaraffa, Antonio Di Florio, Michela Terenzi, Ana Ferreira, Simone Pozzi, Viviana Betti, Matteo Marucci, and Fabio Babiloni. 2020. Involving Hearing, Haptics and Kinesthetics into Non-visual Interaction Concepts for an Augmented Remote Tower Environment. In *Computer Vision, Imaging and Computer Graphics Theory and Applications*, Ana Paula Cláudio, Kadi Bouatouch, Manuela Chessa, Alexis Paljic, Andreas Kerren, Christophe Hurter, Alain Tremeau, and Giovanni Maria Farinella (Eds.). Springer International Publishing, Cham, 73–100.
- [28] Maxime Reynal, Euan Freeman, and Stephen Brewster. 2020. Avoiding Collisions When Interacting with Levitating Particle Displays. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3334480.3382965>
- [29] Maxime Reynal, Jean-Paul Imbert, Pietro Aricò, Jérôme Toupillier, Gianluca Borghini, and Christophe Hurter. 2019. Audio Focus: Interactive spatial sound coupled with haptics to improve sound source location in poor visibility. *International Journal of Human-Computer Studies* 129 (2019), 116 – 128. <https://doi.org/10.1016/j.ijhcs.2019.04.001>
- [30] Diego C. Ruspini, Krasimir Kolarov, and Oussama Khatib. 1997. The Haptic Display of Complex Graphical Environments. In *Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '97)*. ACM Press/Addison-Wesley Publishing Co., USA, 345–352. <https://doi.org/10.1145/258734.258878>
- [31] Majed Samad, Elia Gatti, Anne Hermes, Hrvoje Benko, and Cesare Parise. 2019. Pseudo-Haptic Weight: Changing the Perceived Weight of Virtual Objects By Manipulating Control-Display Ratio. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300550>